

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

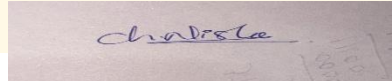
ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0304	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	<i>Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.</i>		
Student Name:	Chodam Nisha	Reg. No.:	192324306
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Code of Conduct

I Chodam Nisha certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

Questions

Q1. A mobile user streams a 3-minute video clip encoded at 1.5 Mbps. How much data does the clip consume?

Given:

Bit rate = 1.5 Mbps

Time = 3 minutes = 180 seconds

Data = $1.5 \times 180 = 270$ Mb

Data in MB = $270 / 8 = 33.75$ MB

Answer: 33.75 MB

Q2. A subscriber has a 30 GB monthly data plan and uses 1.2 GB every day. For how many days will the plan last?

Given:

Monthly data = 30 GB

Daily usage = 1.2 GB

Number of days = $30 / 1.2 = 25$

Answer: 25 days

Q3. A smartphone battery is rated 5000 mAh at a nominal 3.85 V. How much energy does it store?

Given:

Battery capacity = 5000 mAh = 5 Ah

Voltage = 3.85 V

Energy = Ah $\times$ V

Energy = $5 \times 3.85 = 19.25$ Wh

Answer: 19.25 Wh

Q4. A messaging app sends a 2 KB keep-alive message every 30 seconds. How much data does it use in one day?

Number of messages per day = $86400 / 30$

= 2880 messages

Total data = 2880×2 KB
= 5760 KB
= 5.625 MB
Answer: 5.625 MB/day

Q5. Co-channel cells are located with shift parameters $i = 3$ and $j = 2$. What is the resulting cluster size N ?

Formula:
 $N = i^2 + ij + j^2$
 $N = 3^2 + (3 \times 2) + 2^2$
 $N = 9 + 6 + 4$
 $N = 19$
Answer: $N = 19$

Q6. Two co-channel cells of radius 2 km are 13.86 km apart. What cluster size does this reuse distance indicate?

Given:
 $R = 2$ km
 $D = 13.86$ km
Formula:
 $N = (D/R)^2 / 3$
 $N = (13.86/2)^2 / 3$
 $N \approx 16$
Answer: Cluster size = 16

Q7. Find the area covered by one hexagonal cell of radius 3 km.

Given:
 $R = 3$ km
Formula:
 $\text{Area} = (3\sqrt{3}/2)R^2$
 $\text{Area} = (3\sqrt{3}/2)(3^2)$
 $\text{Area} \approx 23.38 \text{ km}^2$
Answer: 23.38 km^2

Q8. A system has 840 duplex channels in total and uses a 12-cell cluster. How many channels does each cell get, and what is the frequency reuse factor?

Given:
Total channels = 840
Cluster size = 12
Channels per cell = $840 / 12$
= 70 channels
Reuse factor = $1 / 12$
= 0.0833
Answer: 70 channels per cell; reuse factor = $1/12$

Q9. A system with 336 channels and a 7-cell cluster covers its service area with 49 cells. How many calls can it carry at the same time?

Channels per cell = $336 / 7$
= 48 channels

$$\text{Total calls} = 48 \times 49$$

$$= 2352$$

Answer: 2352 calls

Q10. A cell of radius 4 km is split into cells of radius 2 km in an area with path-loss exponent 4. By how many dB must the transmit power of the new cells be reduced?

Given:

$$\text{Old radius} = 4 \text{ km}$$

$$\text{New radius} = 2 \text{ km}$$

$$\text{Path-loss exponent} = 4$$

$$\text{Power reduction} = 10n \log_{10}(4/2)$$

$$= 10 \times 4 \times \log_{10}(2)$$

$$\approx 12.04 \text{ dB}$$

Answer: 12.04 dB

Q11. A cell with 84 channels is divided into six 60° sectors with channels shared equally. How many channels does each sector receive?

$$\text{Channels per sector} = 84 / 6$$

$$= 14$$

Answer: 14 channels per sector

Q12. A city of 400 sq. km is served by cells of 8 sq. km each, grouped into 5-cell clusters. How many clusters are needed?

$$\text{Number of cells} = 400 / 8$$

$$= 50 \text{ cells}$$

$$\text{Number of clusters} = 50 / 5$$

$$= 10$$

Answer: 10 clusters

Q13. A 12-cell reuse plan operates with path-loss exponent 4 and six first-tier interferers. Estimate the S/I in dB.

Formula:

$$S/I = (\sqrt[3]{(3N)})^n / i_0$$

$$S/I = (\sqrt[3]{36})^4 / 6$$

$$= 1296 / 6$$

$$= 216$$

$$S/I(\text{dB}) = 10 \log_{10}(216)$$

$$\approx 23.34 \text{ dB}$$

Answer: 23.34 dB

Q14. A 7-cell system operates in an area with path-loss exponent 3 and six first-tier interferers. What S/I does it achieve?

$$S/I = (\sqrt[3]{(3N)})^n / i_0$$

$$S/I = (\sqrt[3]{21})^3 / 6$$

$$\approx 40.15$$

$$S/I(\text{dB}) = 10 \log_{10}(40.15)$$

$$\approx 16.04 \text{ dB}$$

Answer: 16.04 dB

Q15. A receiver needs a C/I of at least 15 dB, and the serving signal is -75 dBm. What is the highest interference level it can tolerate?

$$C/I = C - I$$

$$15 = -75 - I$$

$$I = -90 \text{ dBm}$$

Answer: -90 dBm

Q16. Find the length of a half-wave dipole antenna for a 900 MHz handset.

Given:

$$f = 900 \text{ MHz}$$

$$\text{Wavelength} = c/f$$

$$= 3 \times 10^8 / 900 \times 10^6$$

$$= 0.333 \text{ m}$$

$$\text{Half-wave length} = 0.333 / 2$$

$$= 0.167 \text{ m}$$

Answer: 0.167 m or 16.7 cm

Q17. Find the free-space path loss over 100 m at 2.4 GHz.

Given:

$$\text{Distance} = 100 \text{ m}$$

$$\text{Frequency} = 2.4 \text{ GHz}$$

$$\text{Wavelength} = c/f$$

$$= 0.125 \text{ m}$$

$$\text{FSPL} = 20 \log_{10}(4\pi d/\lambda)$$

$$\text{FSPL} \approx 80.05 \text{ dB}$$

Answer: 80.05 dB

Q18. A base station transmits at 43 dBm with a 15 dBi antenna. The handset antenna gain is 0 dBi and the path loss is 130 dB. What power does the handset receive?

$$\text{Received power} = \text{Transmit power} + \text{Antenna gains} - \text{Path loss}$$

$$P_r = 43 + 15 + 0 - 130$$

$$P_r = -72 \text{ dBm}$$

Answer: -72 dBm

Q19. A car moves at 60 km/h while using a 2600 MHz carrier. Find the maximum Doppler shift.

$$\text{Speed} = 60 \text{ km/h} = 16.67 \text{ m/s}$$

$$\text{Frequency} = 2600 \text{ MHz}$$

$$\text{Doppler shift} = vf/c$$

$$= (16.67 \times 2.6 \times 10^9) / (3 \times 10^8)$$

$$\approx 144.4 \text{ Hz}$$

Answer: 144.4 Hz

Q20. Find the thermal noise floor of a 200 kHz GSM channel at 290 K.

$$\text{Thermal noise power} = -174 + 10 \log_{10}(B)$$

$$B = 200 \text{ kHz}$$

$$\text{Noise} = -174 + 10 \log_{10}(200000)$$

$$\approx -121 \text{ dBm}$$

Answer: -121 dBm

Q21. A GSM TDMA frame lasts 4.615 ms, and a user is given one slot in every frame. How many slots does that user get per second?

$$\text{Slots per second} = 1 / 0.004615$$

$$\approx 216.7$$

Answer: 216.7 slots/second

Q22. A GSM control multiframe contains 51 TDMA frames of 4.615 ms each. How long does one multiframe last?

$$\text{Multiframe duration} = 51 \times 4.615 \text{ ms}$$

$$= 235.365 \text{ ms}$$

Answer: 235.4 ms

Q23. A CDMA link has a processing gain of 21 dB, and the receiver needs an E_b/N_0 of 7 dB. What is the jamming margin?

$$\text{Jamming margin} = \text{Processing gain} - \text{Required } E_b/N_0$$

$$= 21 - 7$$

$$= 14 \text{ dB}$$

Answer: 14 dB

Q24. An OFDMA system uses 15 kHz subcarrier spacing. What is the useful OFDM symbol duration?

$$\text{Symbol duration} = 1 / \text{Subcarrier spacing}$$

$$= 1 / 15000$$

$$= 66.67 \text{ microseconds}$$

Answer: 66.67 microseconds

Q25. Handsets use pure ALOHA on a random access channel with offered load $G = 0.5$. What throughput is achieved?

For pure ALOHA:

$$S = G e^{(-2G)}$$

$$S = 0.5 \times e^{(-1)}$$

$$S \approx 0.184$$

Answer: 0.184 or 18.4%

Q26. A slotted random access channel has 1000 slots per second, and 200 access attempts arrive each second. How many attempts succeed per second?

Offered load:

$$G = 200 / 1000$$

$$= 0.2$$

Throughput:

$$S = G e^{(-G)}$$

$$S = 0.2 \times e^{(-0.2)}$$

$$\approx 0.1637$$

$$\text{Successful attempts} = 1000 \times 0.1637$$

Answer: Approximately 164 attempts/second

Q27. A paging channel offers 40 paging blocks per second, and each block carries 2 pages. A location area needs 50 pages per second. What is the paging channel utilisation?

$$\text{Capacity} = 40 \times 2$$

$$= 80 \text{ pages/second}$$

$$\text{Utilisation} = 50/80 \times 100$$

Answer: 62.5%

Q28. A BTS has 4 TRXs with 8 time slots each, and 2 slots are reserved for signalling. How many traffic channels are available?

$$\text{Traffic channels} = 4 \times (8 - 2)$$

$$= 4 \times 6$$

$$= 24$$

Answer: 24 traffic channels

Q29. A BSC can handle 512 TRXs. The network has 400 BTSs with 3 TRXs each. How many BSCs are needed?

$$\text{Total TRXs} = 400 \times 3$$

$$= 1200 \text{ TRXs}$$

$$\text{BSCs required} = 1200 / 512$$

$$= 2.34$$

Since partial BSC is not possible:

Answer: 3 BSCs

Q30. An E1 link has 32 time slots of 64 kbps, of which 2 are used for synchronisation and signalling. Each remaining slot carries four 16 kbps traffic channels. How many traffic channels does the E1 carry?

$$\text{Available slots} = 32 - 2$$

$$= 30$$

$$\text{Traffic channels} = 30 \times 4$$

Answer: 120 traffic channels

Q31. An EIR stores an 8-byte record for each registered IMEI. How much storage is needed for 20 million devices?

$$\text{Storage} = 20,000,000 \times 8 \text{ bytes}$$

$$= 160,000,000 \text{ bytes}$$

Answer: 160 MB

Q32. A VLR requests 5 authentication triplets from the AuC for each of 100,000 subscribers per hour. How many triplets must the AuC generate per second?

$$\text{Triplets per hour} = 5 \times 100,000$$

$$= 500,000$$

$$\text{Triplets per second} = 500,000 / 3600$$

$$\approx 138.9$$

Answer: Approximately 139 triplets/second

Q33. A TRX carries 8 full-rate voice channels. If it switches to half-rate coding, how many users can it carry, and what is the capacity gain?

$$\text{Full-rate users} = 8$$

Half-rate doubles the capacity:

$$\text{Users} = 8 \times 2$$

$$= 16$$

$$\text{Capacity gain} = 100\%$$

Answer: 16 users; 100% capacity gain

Q34. A TCP segment carries 536 bytes of data with 20-byte TCP and 20-byte IP headers. What is the transmission efficiency?

$$\text{Total size} = 536 + 20 + 20$$

$$= 576 \text{ bytes}$$

$$\text{Efficiency} = 536 / 576 \times 100$$

$$\approx 93.06\%$$

Answer: 93.06%

Q35. A 1500-byte IP packet is segmented into radio link blocks that each carry 40 bytes of payload. How many blocks are needed?

$$\text{Number of blocks} = 1500 / 40$$

$$= 37.5$$

Since partial block is not possible:

Answer: 38 blocks

Q36. A radio link has a 10% block error rate, and every failed block is retransmitted until it succeeds. What is the average number of transmissions per block?

$$\text{Success probability} = 1 - 0.10$$

$$= 0.90$$

$$\text{Average transmissions} = 1 / 0.90$$

Answer: 1.11 transmissions per block

Q37. An SMS carries a 140-byte payload using 7-bit character coding. How many characters can one SMS hold?

$$140 \text{ bytes} = 140 \times 8$$

$$= 1120 \text{ bits}$$

$$\text{Characters} = 1120 / 7$$

Answer: 160 characters

Q38. A PLMN uses a 16-bit Location Area Code. What is the maximum number of location areas it can define?

$$\text{Maximum values} = 2^{16}$$

$$= 65,536$$

Answer: 65,536 location areas

Q39. A Cell Identity field is 16 bits long within each location area, and the LAC is also 16 bits. How many unique cells can a PLMN identify?

$$\text{Total bits} = 16 + 16$$

$$= 32 \text{ bits}$$

$$\text{Unique cells} = 2^{32}$$

Answer: 4,294,967,296 unique cells

Q40. An SS7 signalling link runs at 64 kbps and is engineered for 40% load. If each MAP message is 150 bytes, how many messages can the link carry per second?

$$\text{Usable rate} = 64,000 \times 0.40$$

$$= 25,600 \text{ bps}$$

$$\text{Message size} = 150 \times 8$$

$$= 1200 \text{ bits}$$

$$\text{Messages/second} = 25,600 / 1200$$

Answer: Approximately 21.33 messages/second

Q41. A data packet crosses 4 core-network nodes. Each node adds 2 ms of processing, and each hop adds 1 ms of transmission. What is the one-way core delay?

$$\text{Processing delay} = 4 \times 2$$

$$= 8 \text{ ms}$$

$$\text{Transmission delay} = 4 \times 1$$

$$= 4 \text{ ms}$$

$$\text{Total delay} = 8 + 4$$

Answer: 12 ms

Q42. An average call lasts 120 seconds, and a user moves into a new cell every 60 seconds. How many handoffs occur during an average call?

$$\text{Call duration} = 120 \text{ seconds}$$

$$\text{Cell change interval} = 60 \text{ seconds}$$

At 60 seconds, the user changes cell once.

Answer: 1 handoff

Q43. A pedestrian walks at 1.5 m/s through a microcell 500 m across. How long does the user stay in the cell?

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$= 500 / 1.5$$

$$= 333.33 \text{ seconds}$$

Answer: 333.33 seconds or 5.56 minutes

Q44. Out of 10,000 handoff attempts, 9700 succeed. What are the handoff success rate and failure rate?

$$\text{Success rate} = 9700 / 10000 \times 100$$

$$= 97\%$$

$$\text{Failure rate} = 100 - 97$$

$$= 3\%$$

Answer: Success rate = 97%; Failure rate = 3%

Q45. Of 50 handoffs recorded on a road, 12 returned to the original cell within 5 seconds. What is the ping-pong handoff rate?

Given:

$$\text{Total handoffs} = 50$$

$$\text{Ping-pong handoffs} = 12$$

$$\text{Ping-pong handoff rate} = (12 / 50) \times 100$$

$$= 24\%$$

Q46. A visited MSC has a pool of 5000 roaming numbers (MSRNs). Each is held for 30 seconds during call setup. How many roaming call setups can it handle per hour?

$$\text{Setups per 30 seconds} = 5000$$

Number of 30-second intervals per hour:

$$3600 / 30 = 120$$

$$\text{Total setups} = 5000 \times 120$$

Answer: 600,000 call setups/hour

Q47. A location area has 10 cells. A user receives 6 calls per hour, each paged in all 10 cells, and makes 2 location updates per hour, each costing 5 messages. What is the total signalling per hour for this user?

Paging messages = 6×10
= 60 messages/hour
Location update messages = 2×5
= 10 messages/hour
Total signalling = $60 + 10$
Answer: 70 messages/hour

Q48. A cell receives 300 calls per hour with an average holding time of 3 minutes. What is the offered traffic in Erlangs?

Holding time = $3/60$
= 0.05 hour
Traffic = Calls/hour $\times$ Holding time
= 300×0.05
Answer: 15 Erlangs

Q49. Of 50,000 call attempts, 48,500 are set up successfully. What is the call setup success rate?

Given:
Total call attempts = 50,000
Successful calls = 48,500
Call setup success rate = $(48,500 / 50,000) \times 100$
= 97%

Q50. A cellular network is out of service for 26.28 hours in a year of 8760 hours. What is its availability?

Availability = $(\text{Total time} - \text{Downtime}) / \text{Total time} \times 100$
= $(8760 - 26.28) / 8760 \times 100$
Answer: 99.7%